Lab Assignment - 10

Phylogenetic Tree Construction and Weight Matrix

BT3040 - Bioinformatics

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Question 1

The answers to the questions are presented section-wise, in accordance with the structure of the assignment question set.

Part (i, ii)

Multiple Sequence Alignment was performed on both "tim.dat" and "tim-hemo.dat" using MAFFT and the output was saved as "work1" and "work2" in `Phylip4` format respectively.

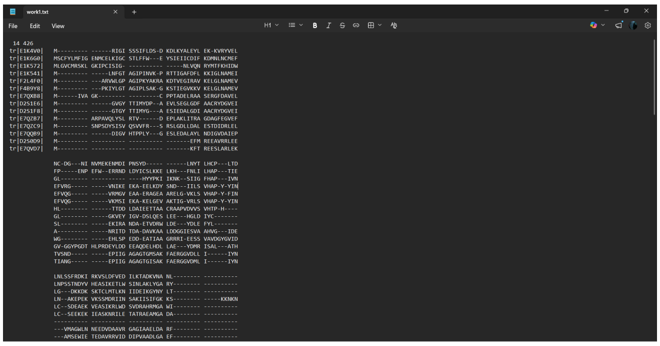

Figure 1.ii.1: MSA output (tim.dat)

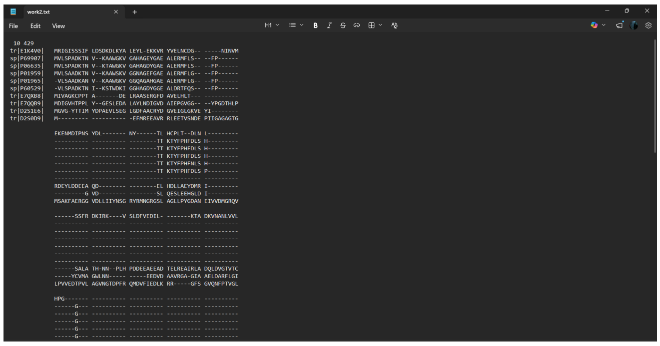

Figure 1.ii.2: MSA output (tim-hemo.dat)

Part (iii)

The PHYLIP software was successfully downloaded and installed on the Windows system by following the provided instructions.

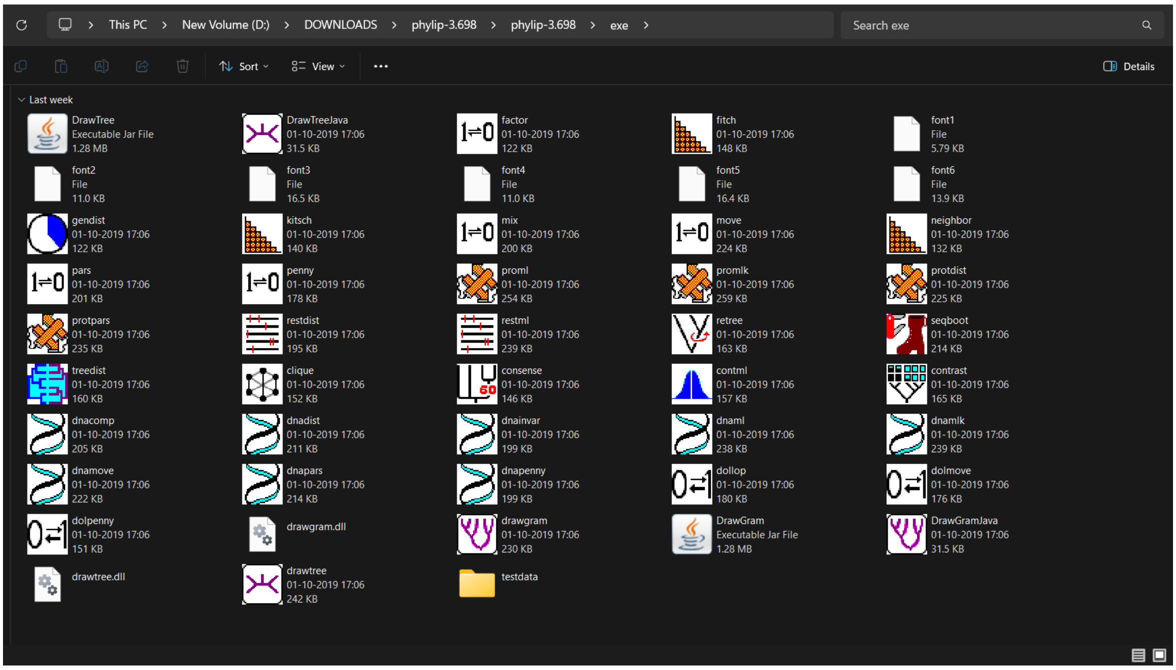

Figure 1.iii.1: PHYLIP's executable programs

Part (iv)

Bootstrapping was performed using the `seqboot` program, with the number of replicates set to 10 and the random seed fixed at 5 (to ensure reproducibility of the results).The output was written to files "outfile_1_seqboot" and "outfile_2_seqboot" respectively. The program outputs are shown below:

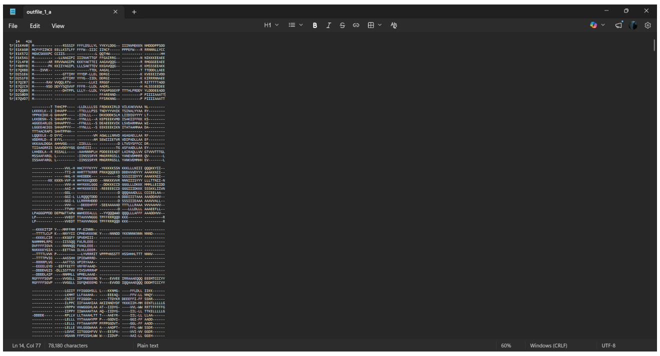

Figure 1.iv.1: `seqboot` Output for set-1

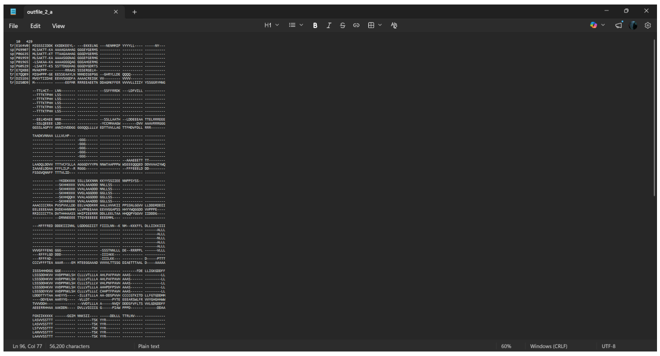

Figure 1.iv.2: `seqboot` Output for set-2

Part (v)

The outfiles produced by `seqboot` are used as the input for the `proml` program to compute the phylogenetic tree based on maximum likelihood measures.

The output of `proml` was written to files "outfile/tree_1_proml" and "outfile/tree_2_proml" respectively. The program outputs are shown below:

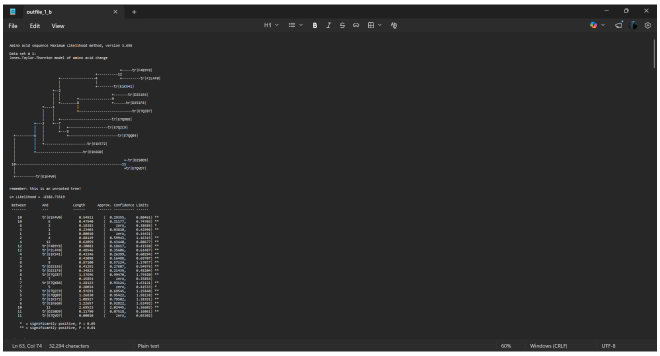

Figure 1.v.1: `proml` Output for set-1

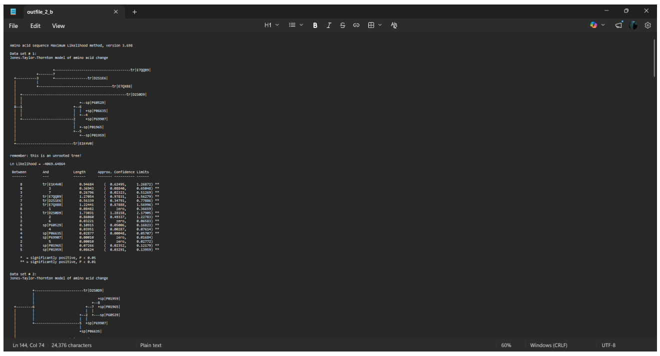

Figure 1.v.2: `proml` Output for set-2

Part (vi, vii)

The outtree files obtained from `proml` are fed into the `consense` program to obtain the consensus tree. The output files of the `consense` program, "outtree_1_consense_1.txt" and "outtree_2_consense_1.txt" respectively were converted into the NWK file format and the phylogenetic tree was visualized using the `MEGA-X` program as shown below:

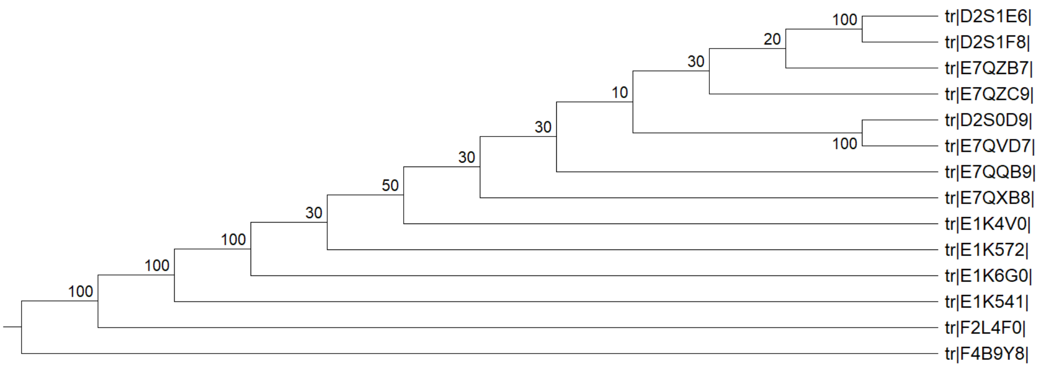

Figure 1.vii.1: Set-1, Maximum Likelihood based Phylogenetic tree

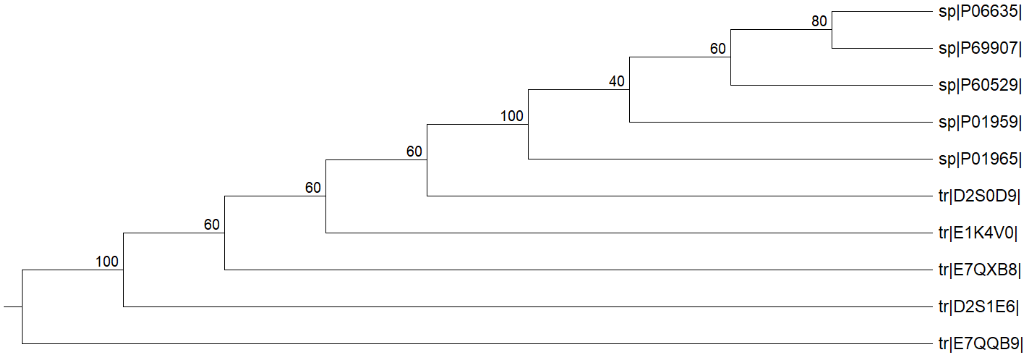

Figure 1.vii.2: Set-2, Maximum Likelihood based Phylogenetic tree

Part (viii)

To execute NJ & UPGMA, the outfiles from seqboot ("outfile_1_seqboot" and "outfile_2_seqboot") were fed into the `protdist` program, with number of datsets set to 10. The output obtained is shown below:

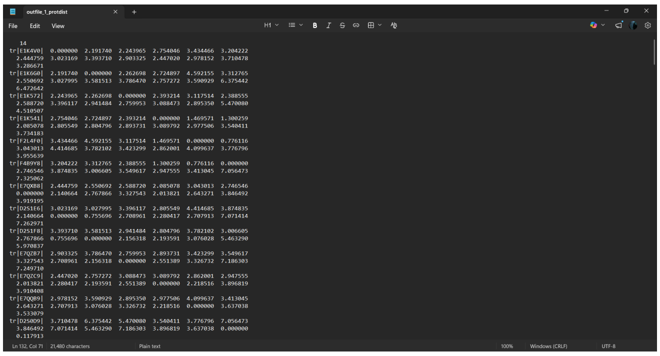

Figure 1.viii.1: `protdist` Output for set-1

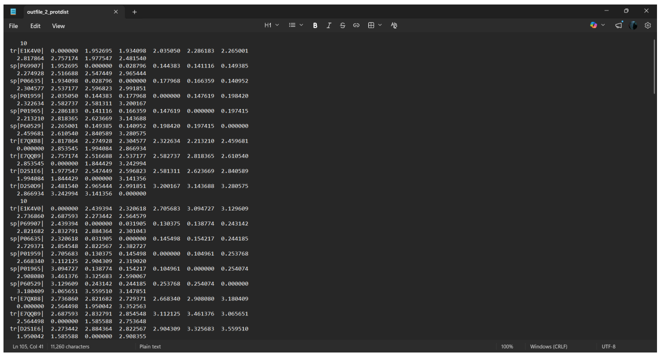

Figure 1.viii.2: `protdist` Output for set-2

The output from protdist is fed into the `neighbor` program, and the output was written to the files "outfile/tree_1_neighbor" and "outfile/tree_2_neighbor" respectively. The program outputs are shown below:

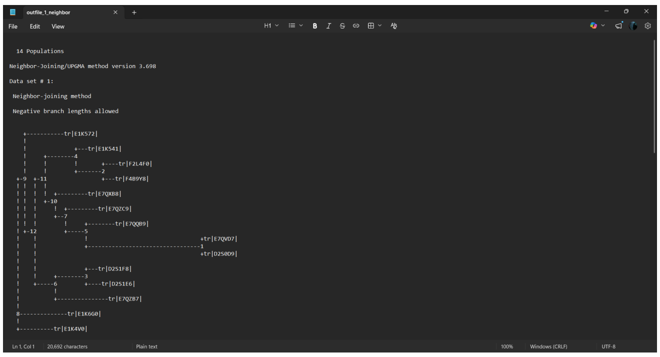

Figure 1.viii.3: `neighbor` Output for set-1

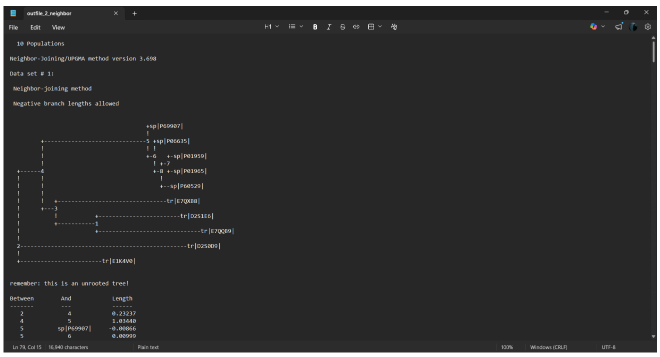

Figure 1.viii.4: `neighbor` Output for set-2

The outtree files obtained from `neighbor` are fed into the `consense` program to obtain the consensus tree. The output files of the `consense` program, "outtree_1_consense_2.txt" and "outtree_2_consense_2.txt" respectively were converted into the NWK file format and the phylogenetic tree was visualized using the `MEGA-X` program as shown below:

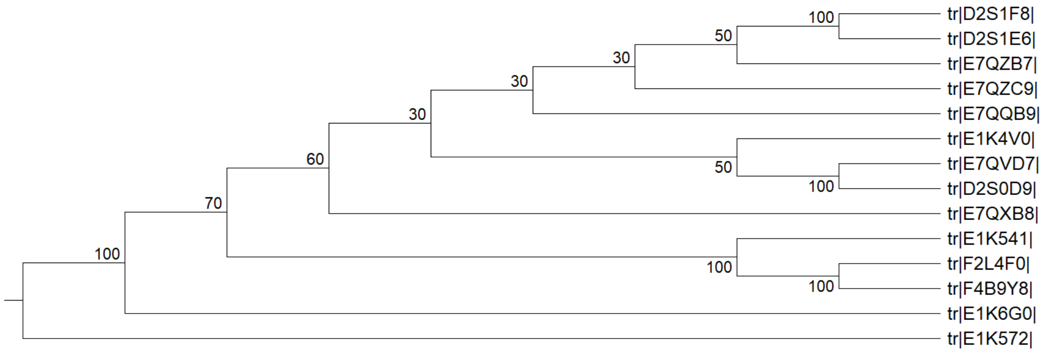

Figure 1.viii.5: Set-1, NJ & UPGMA based Phylogenetic tree

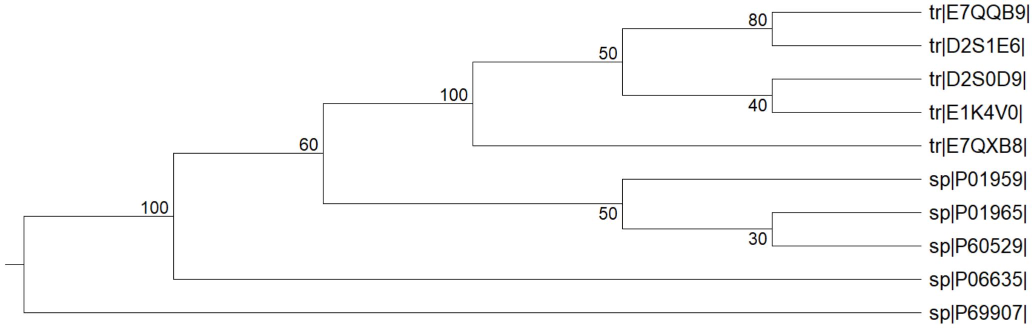

Figure 1.viii.6: Set-2, NJ & UPGMA based Phylogenetic tree

Question 2

The code to calculate the create the weight matrix for the given sequences is shown below:

```
# Question 2

seq1 = "MVLSPADKTNVKGKVGAHAGEYGAAAW"
seq2 = "MKRLPADPPCVKGKVKAKAGDYGATTW"
seq3 = "MALSAADKTNVKSKVGGHAGEYGAATS"
seq4 = "MVLSAADKTNVKSKAGGNAGEWWAAAW"
seq5 = "MVLSAADKTNVKSKVLANAGEFGAAAW"
seq6 = "ALLPIRTTYHKKCASGHIPEEKDLNNV"
seq7 = "DEASSLKGHHIKKLEADALLIPLSASS"

sequences = {}
sequences["sequence_1"] = seq1
sequences["sequence_2"] = seq2
sequences["sequence_3"] = seq3
sequences["sequence_4"] = seq4
sequences["sequence_5"] = seq5
sequences["sequence_6"] = seq6
sequences["sequence_7"] = seq7

AAs = "ACDEFGHIKLMNPQRSTVWY"
n = len(AAs)
m =len(seq1)
```

```
def weight_mat(sequences):
  p = 1/20
  arr = np.zeros((n, m))

  for position in range(m):
    residues = {}

    for AA in AAs:
      residues[AA] = 0

    for seq_id in sequences.keys():
      residues[sequences[seq_id][position]] += 1

    for residue in range(len(AAs)):
      arr[residue][position] = math.log(
      (residues[AAs[residue]] + p)/((len(sequences)+1)*p)
      )

  df = pd.DataFrame(
    arr,
    index=list(AAs),
    columns=range(1, m + 1)
  )
  df = df.round(2)
  pd.set_option('display.max_columns', None)
  pd.set_option('display.width', None)
  pd.set_option('display.expand_frame_repr', False)

  print(df)

weight_mat(sequences)
```

The output of the code is given below:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
A	0.97	0.97	0.97	-2.08	2.03	2.54	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	0.97	0.97	2.03	0.97	2.54	-2.08	-2.08	-2.08	-2.08	2.54	2.54	2.03	-2.08
C	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
D	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	2.54	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	0.97	-2.08	0.97	-2.08	-2.08	-2.08	-2.08
E	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	0.97	2.54	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
F	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08
G	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	1.63	-2.08	-2.08	2.32	1.63	-2.08	-2.08	2.54	-2.08	-2.08	2.32	-2.08	-2.08	-2.08	-2.08
H	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	1.63	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	1.63	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
I	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
K	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	0.97	2.32	-2.08	-2.08	0.97	2.87	0.97	2.54	-2.08	0.97	-2.08	0.97	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08
L	-2.08	0.97	2.54	0.97	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	0.97	-2.08	-2.08	0.97	0.97	-2.08	-2.08	0.97	0.97	-2.08	-2.08	-2.08
M	2.54	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
N	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	2.32	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	1.63	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	0.97	-2.08
P	-2.08	-2.08	-2.08	0.97	1.63	-2.08	-2.08	0.97	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08
Q	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
R	-2.08	-2.08	0.97	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08
S	-2.08	-2.08	-2.08	2.54	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	2.03	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	0.97	1.63
T	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	0.97	2.32	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	1.63	-2.08
V	-2.08	2.03	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	2.54	-2.08	-2.08	-2.08	2.32	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97
W	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	0.97	-2.08	-2.08	-2.08	2.32
Y	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	0.97	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	-2.08	2.03	-2.08	-2.08	-2.08	-2.08	-2.08

Figure 2.1: Weight Matrix

Link to the google drive containing the code and the result files from the different tools used in this assignment: `https://drive.google.com/drive/u/0/folders/1n83f1YqpkEj2KbCpG9IaPEL-OO0aK7u0`